Note 2.1 The Determinant of a Matrix

Determinant for Simple Matrices

- 1×1 **Matrices:**
  For $A = (a)$, then we define

$$\det(A) = a$$

  then A will be nonsingular if and only if $\det(\mathrm{A}) \neq 0$.
- 2×2 **Matrices**
  For

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

  then we define

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}$$

  then A is nonsingular if and only if $\det(A) \neq 0$.

Notation

- **Representation:**
  For a matrix

$$A = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

  then

$$\begin{vmatrix} a_{11} & a_{21} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

  represents the determinant of A.
- **Definition of minor and cofactor:**
  Let $A = (a_{ij})$ be $n \times n$ matrix, and let M_{ij} denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting the ith row and the jth column. Then the determinant of M_{ij} is called the

minor of a_{ij}. And we define the **cofactor** A_{ij} of a_{ij} by

$$A_{ij} = (-1)^{i+j}\det(M_{ij})$$

- **Definition of Determinant:**
  The determinant of an $n \times n$ matrix A, denoted by $\det(A)$, is a scalar to judge whether the matrix is singular and associated with the matrix A. It is defined as

  $$\det(A) = \begin{cases} a_{11}, \text{if } n = 1 \\ a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n}, \text{if } n > 1 \end{cases}$$

  where

  $$A_{1j} = (-1)^{1+j}\det(M_{1j}), j = 1, \cdots, n$$

  are the cofactors associated with the entries in the first row of A.
- **Understanding of Determinant:**
  If a determinant is not zero, it represents the sketched space of the vectors for the corresponding matrix, or the factor of the n-dimensional space's linear transformation for the corresponding matrix. Its sign represents whether the direction among the vectors change.
  If a determinant is zero, it represents the sketched space of the vectors for the corresponding matrix is zero, or the transformation squishes the space into a smaller dimension. And the corresponding matrix is singular since after the transformation there is no linear transformation can turn it into the original space and it must map multiple vectors.
- **Theorem 2.1.1:**
  If A is an $n \times n$ matrix with $n \geq 2$, then $\det(A)$ can be expressed as a cofactor expansion of any row or column of A, that is

  $$\begin{aligned} \det(A) &= a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} \\ &= a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} \end{aligned}$$

  for $i = 1, \cdots, n$ and $j = 1, \cdots, n$. ^f2afdf
  Understanding:
  The recursion definition of determinant actually is divide the higher-dimensional space into the linear combination of lower-dimensional space. The selected line is the direction we need to map the higher-dimensional space into. The size of the lower-dimensional space is the corresponding cofactor and its height is the value of the selected entry.
  - Example:
    To evaluate

    $$\begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

we would expand down the first column since the first three entries are zero, leaving

$$-2\begin{vmatrix}2 & 3 & 0\\ 4 & 5 & 0\\ 1 & 0 & 3\end{vmatrix} = -2\cdot 3\cdot\begin{vmatrix}2 & 3\\ 4 & 5\end{vmatrix} = 12$$

- **Theorem 2.1.2:**
  If A is an $n\times n$ matrix, then $\det(A^T) = \det(A)$.
- **Theorem 2.1.3:**
  If A is an $n\times n$ **triangular** matrix, then the determinant of A equals the product of the diagonal elements of A.
- **Theorem 2.1.4:**
  Let A be an $n\times n$ matrix, If
  - A has a row or column consisting entirely of zeros
  - or A has two identical rows or columns
    Then $\det(A) = 0$.

Section 2.1 Exercise

- Question 10th
  When $n = 1$, then the matrix $A = \begin{pmatrix}a & b\\ a & b\end{pmatrix}$, $\det(A) = ab - ab = 0$. Then suppose the result holds for $(k+1)\times(k+1)$ matrix. When the matrix is $(k+2)\times(k+2)$ and the ith row and j th row are the identical. We expand $\det(A)$ by factors along the mth row where $m \neq i$ and $m \neq j$. Then

  $$\det(A) = a_{m1}\det(M_{m1}) + a_{m2}\det(M_{m2}) + \cdots + a_{m,k+2}\det(M_{m,k+2})$$

  where $M_{ms}, 1\ \ s\ \ k+2$ is a $(k+1)\times(k+1)$ matrix having two identical rows. So $\det(M_{ms}) = 0$, $\det(A) = 0$.